

Spike: 8**Title:** Soldier on Patrol**Author:** Edward Herrod, 104541862**Goals / deliverables:**

The expected outcomes of this spike include a working simulation of a patrolling soldier which is independently able to shoot at a stationary target. Its behaviour should be dictated by a layered FSM architecture which includes high-level behaviour (patrolling or attacking) and low-level behaviour (shooting, reloading, fleeing, strafing, seeking). This spike demonstration also includes multiple attacking modes which are dependent on the type of weapon the agent has equipped.

Technologies, Tools, and Resources used:

List of information needed by someone trying to reproduce this work:

- Visual Studio Code
- Python 3.x
- Pyglet 2.1.x
- Git
- Github (for installation of code)
- A working installation of Spike 7 to serve as a code base

Tasks undertaken:

- Download and install Visual Studio Code
- Download and install the latest version of Python
- Install the latest version of Pyglet (pip install --upgrade --user piglet)
- Import the spike containing the base code for this spike (Task 15)
- Update the target agent to have hitpoints and die when they reach 0
- Create a layered FSM file containing the states for the agent
- Implement modular state machines – High-level states Patrol and Attack and Low-level states Flee, Strafe, Seek, Shoot and Reload
- Give each weapon an ammo count and reload timer
- Have the agent follow these states depending on the weapon it's holding

What we found out:

State transitions are caused by a combination of factors, including:

- The existence of a target
- The agent's distance to the target if it exists
- Whether or not the agent has ammo

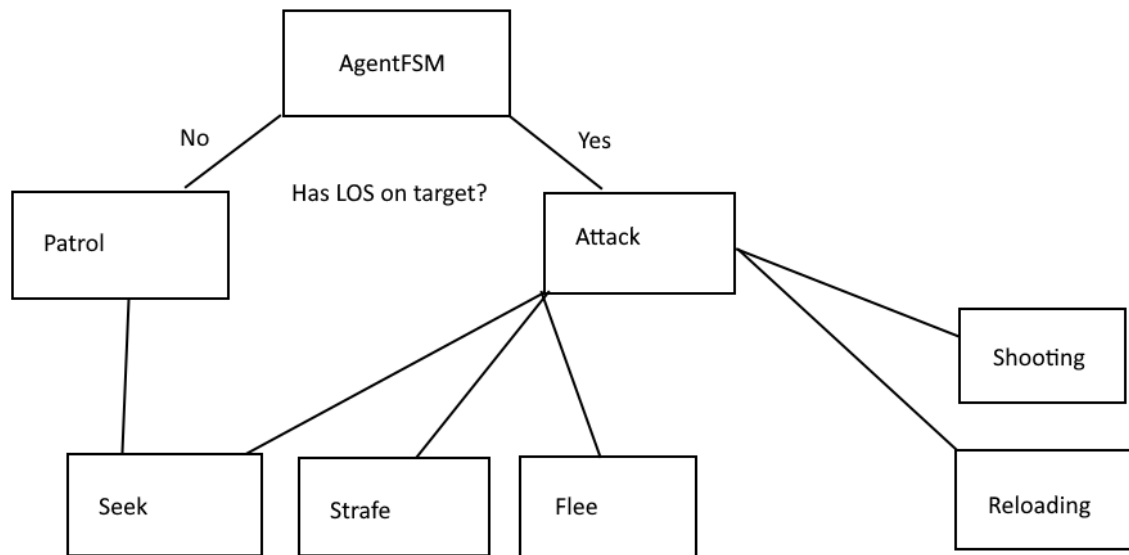


Figure 1: A diagram containing the different FSM states for the agent.



Figure 2: The agent stops along its patrol path to shoot when it is holding a rifle. Note the ammo count in the top-right.

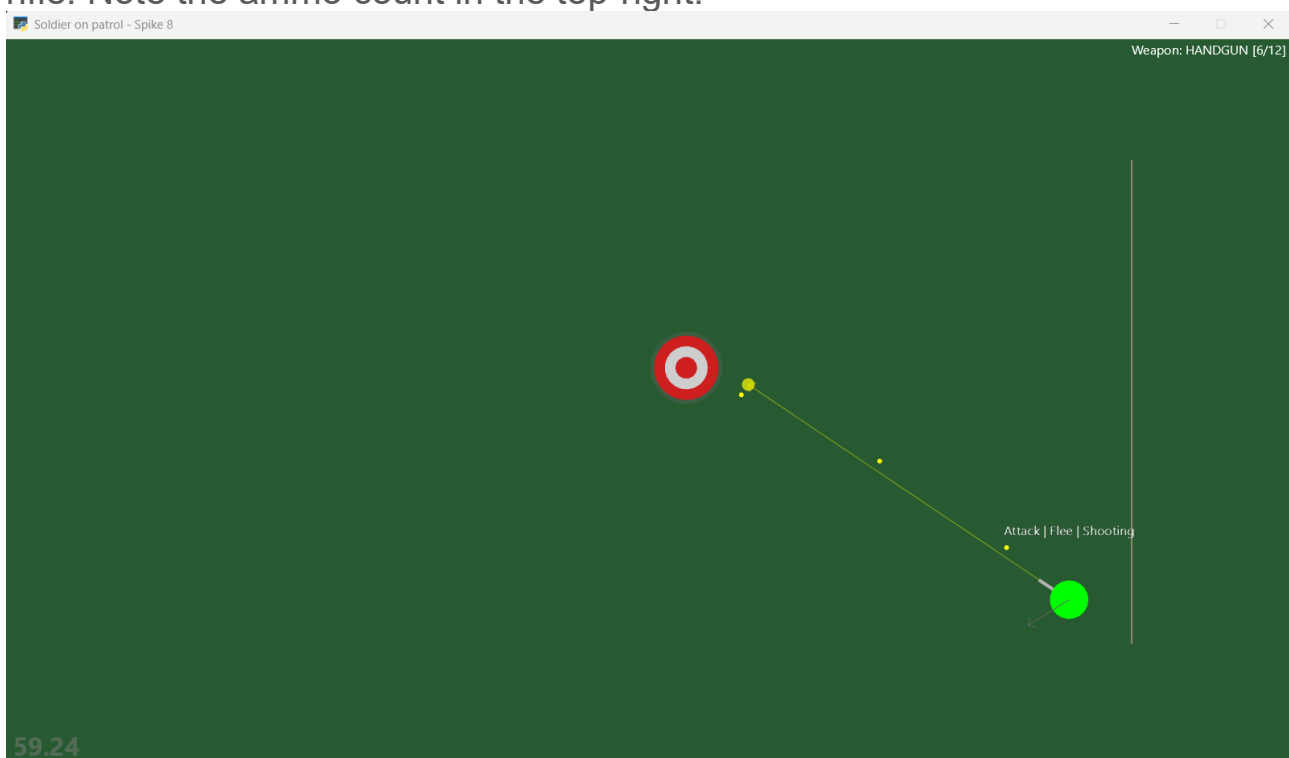


Figure 3: The agent flees from the target when it is holding a handgun.

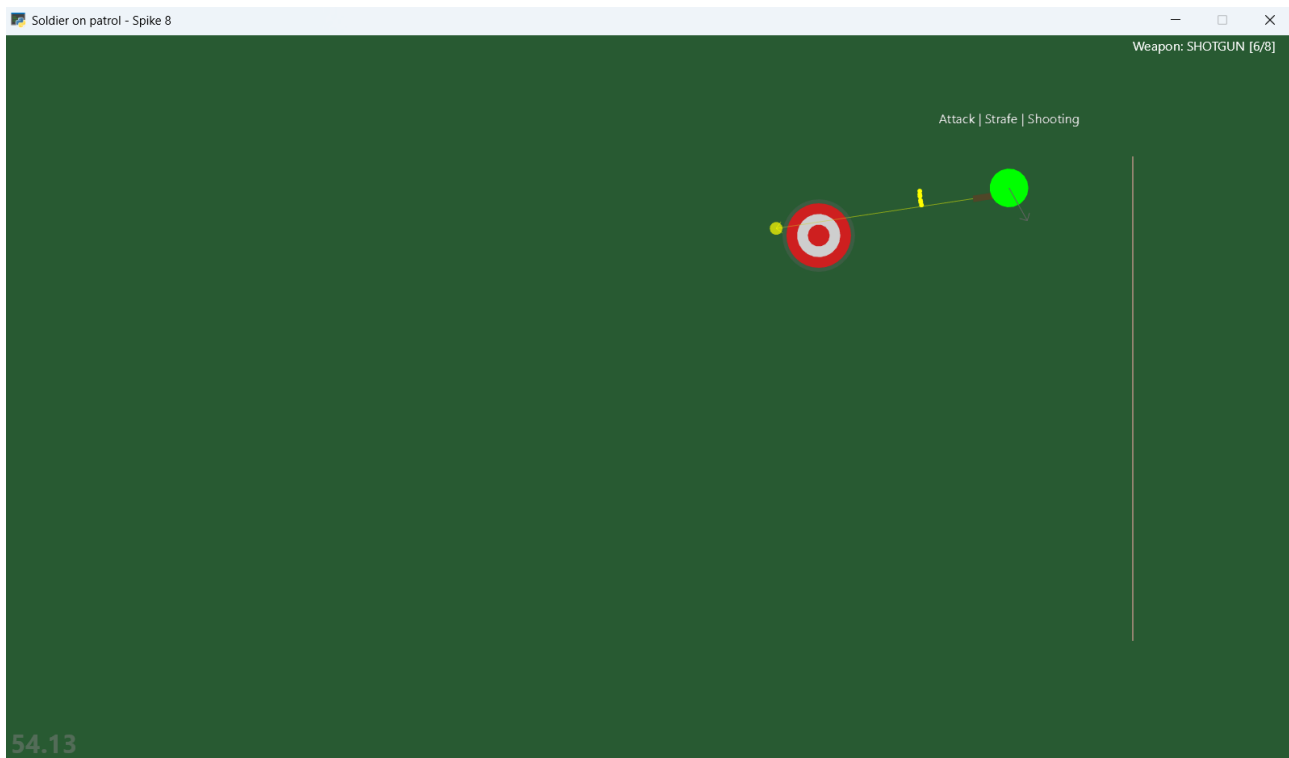


Figure 4: The agent strafes the target at close range when it has a shotgun.

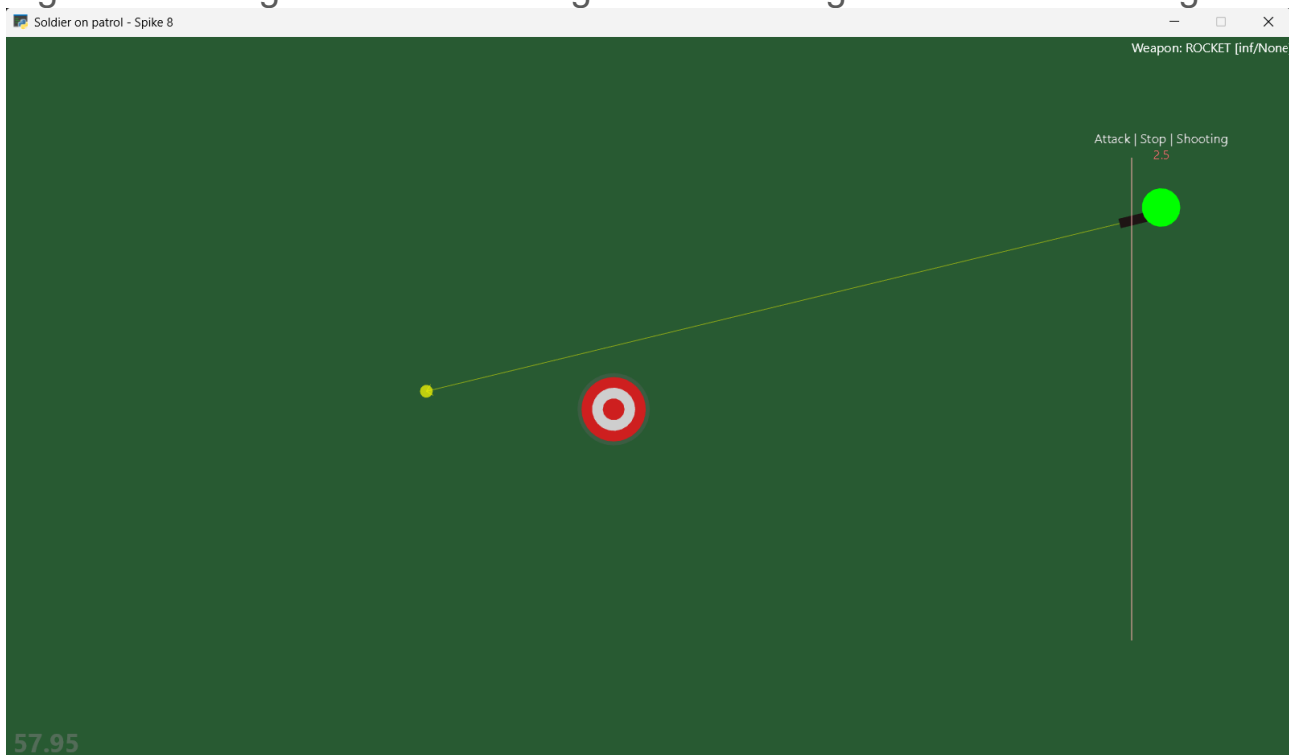


Figure 5: The agent flees to a set distance from the target then locks on (see the red lock-on timer) with its rocket launcher (which has infinite ammo, given the long cooldown) before shooting a one-hit kill explosive projectile.

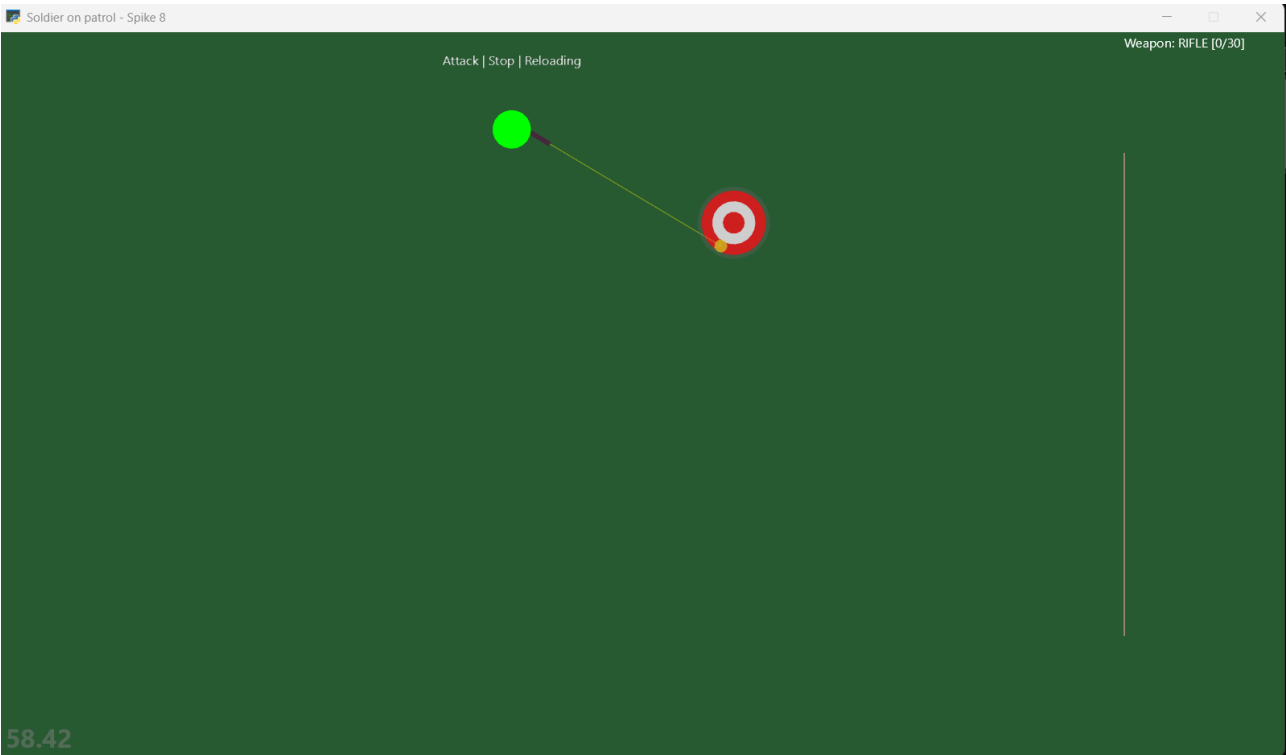


Figure 6: The agent reloads on its own when it runs out of bullets.